

Akash Pandey | Curriculum Vitae

PhD Candidate, Mechanical Engineering – Northwestern University, Illinois 60208

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My Webpage • Github in LinkedIn

Education

Northwestern University

PhD, Mechanical Engineering

Accelerating protein design through explainable deep-learning models

Advisor : Dr. Sinan Keten and Dr. Wei Chen

CGPA : 3.977/4

USA

2021–present

Indian Institute of Technology (IIT) Madras

Masters (by research), Applied Mechanics

Study of the fatigue behavior of piezoelectric composite material

Advisor : Dr. A. Arockiarajan

CGPA : 9.4/10

India

2014–2017

Madras Institute of Technology, Anna University

Bachelor of Engineering, Automobile Engineering

CGPA : 8.78/10

India

2010–2014

Fellowships & Awards

March, 2025: 250 USD travel grant for ICLR'25.

Sept, 2023: Recipient of Predictive Science and Engineering Design (PSED) fellowship at Northwestern University

May, 2023: Top performer in the Association for Computing Machinery (ACM) Multimedia 2023 Computational Paralinguistics Challenge (ComParE).

Feb, 2023: Among top 3 performers in e-Prevention: Person Identification and Relapse Detection from Continuous Recordings of Biosignals Challenge in IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP'23).

Sept, 2021 - June, 2022: Walter P. Murphy Fellowship at Northwestern University

July, 2014 - April, 2017: Awarded Research Fellowship by Government of India

Skills

ML Core: Transformer Architectures, Explainable AI (XAI), Interpretable Deep Learning, Large Language Model (LLM) Fine-Tuning, Machine Learning for Proteomics and Biomedical Applications, Temporal Attribution in Time-Series, Recommender Systems, Active learning, Gaussian Processes, Bayesian Optimization

Programming: Python (PyTorch, Captum, scikit-learn, pandas, NumPy, Jupyter Notebooks), L^AT_EX

Software & Platforms: AWS, MATLAB, Git, LAMMPS, ABAQUS, Siemens NX

Additional Skills: Finite Element Analysis (FEA), Embedded Systems (Raspberry Pi Pico), Numerical Methods

Research Experience

Northwestern University

(2025) Interpretable Model for temporal attribution in time-series data. Developed TimeSliver, an interpretable deep learning model that integrates raw and symbolically binned time-series data to capture temporal interactions and compute temporal attribution scores, achieving a 10% performance improvement over state-of-the-art methods.

(2025) Interpretable Model for monomeric attribution in protein sequences. Developed an interpretable deep learning model, COLOR, that transforms higher-dimensional protein sequences into a lower-dimensional interpretable representation to estimate the contribution of each monomer to a given property. COLOR achieves 22% higher explainability than the existing gradient- and attention-based methods.

(2025) EMG-to-Text conversion with LLMs. Developed an L1aMA 3-based model to convert surface electromyography (EMG) signals, which capture muscle activations, into speech. On a closed vocabulary task, our model achieves approximately 20% lower word error rate (WER) compared to specialized models.

(2024) Predictive model for spider silk's mechanical property. Developed an interpretable feature-based deep learning framework to predict the properties of spider silk and identify important motifs in a data-constrained setting. We showed that using the B-factor as a motif descriptor improves prediction performance by 15% compared to traditional descriptors such as hydrophobicity, charge, and others.

(2023) B-factor prediction in proteins. Developed a many-to-many LSTM model to predict the B-factor (atomic flexibility) of alpha-carbon atoms in proteins, achieving a 30% improvement over the CNN-based state-of-the-art model. Analysis revealed that atoms within 15 Å contribute most significantly to B-factor values.

(2023) Audio-based emotion prediction. As part of an ACM Multimedia Challenge, we developed an emotion prediction model based on an audio foundation model. We found that using HuBERT-Large as the audio foundation significantly improved performance by 4%.

(2023) Person identification based on the biosignals. As part of the ICASSP'23 Challenge, we developed a wav2vec-based deep learning model to identify individuals based on their biosignals, securing 3rd place. We employed a late fusion strategy to effectively handle both time-varying and static features.

IIT Madras

Developed a finite element method and experimental technique to study the fatigue behavior of piezoelectric composite material under mechanical, electrical, and thermal load.

Professional Experience

Capital One: Data Science PhD Intern (San Jose, CA, June 2025 - Aug 2025)

Applied and benchmarked 10 explainable methods to quantify ad influence on user decisions, and built a multi-GPU pipeline for a Transformer-based recommendation model, delivering a minimum viable product and accelerating readiness for production deployment.

Capital One: Data Science PhD Intern (San Jose, CA, June 2024 - Aug 2024)

Applied permutation- and perturbation-based attribution methods to a Transformer-based recommendation model, identifying the top 5 of 40 key features and enhancing model interpretability and trust.

Rolls Royce: Advanced Engineer & Infosys Ltd: Engineering Lead (India, Jan 2019 - Aug 2021)

Developed and deployed computational algorithms to estimate and monitor the service life of aero-engine components, enabling real-time tracking of component health in operation.

Rolls Royce: Global Engineering Graduate (India and UK, July 2017 - Dec 2018)

Rotated across four teams in aero-engine design, analysis, manufacturing, and testing, gaining broad end-to-end engineering exposure. Selected for Rolls-Royce's accelerated leadership program upon completion of the graduate program.

Publications

Journal Articles

2025: Akash Pandey, Wei Chen, and Sinan Keten. COLOR: A compositional linear operation-based representation of protein sequences for identification of monomer contributions to properties. *Journal of Chemical Information and Modeling*. ACS Publications, 2025.

2024: Akash Pandey, Wei Chen, and Sinan Keten. Sequence-based data-constrained deep learning framework to predict spider dragline mechanical properties. *Communications Materials*, volume 5, page 83. Nature Publishing Group UK London, 2024.

2023: Akash Pandey, Elaine Liu, Jacob Graham, Wei Chen, and Sinan Keten. B-factor prediction in proteins using a sequence-based deep learning model. *Patterns*, volume 4. Elsevier, 2023.

2017: Akash Pandey and A. Arockiarajan. Performance studies on macro fiber composite (MFC) under thermal condition using kirchhoff and mindlin plate theories. *International Journal of Mechanical Sciences*, volume 130, pages 416–425, 2017.

2017: Akash Pandey and A. Arockiarajan. Fatigue study on the sensor performance of macro fiber composite (MFC): Theoretical and experimental approach. *Composite Structures*, volume 174, pages 301–318, 2017.

2017: Akash Pandey and A Arockiarajan. Fatigue study on the actuation performance of macro fiber composite (MFC): theoretical and experimental approach. *Smart Materials and Structures*, volume 26, page 035018. IOP Publishing, 2017.

2017: Akash Pandey and A. Arockiarajan. An experimental and theoretical fatigue study on macro fiber composite (MFC) under thermo-mechanical loadings. *European Journal of Mechanics - A/Solids*, volume 66, pages 26–44, 2017.

2016: Akash Pandey and A. Arockiarajan. Actuation performance of macro-fiber composite (MFC): Modeling and experimental studies. *Sensors and Actuators A: Physical*, volume 248, pages 114–129, 2016.

Conferences and Talks (* indicates equal contribution)

2025: Akash Pandey, Payal Mohapatra, Wei Chen, Qi Zhu, and Sinan Keten. TimeSliver : Interpretable time-series classification using symbolic-linear representation. *In Under review*, 2025.

2025: Akash Pandey, Wei Chen, and Sinan Keten. COLOR: A compositional linear operation-based representation of protein sequences for identification of monomer contributions to properties. *MLGenX Workshop at International Conference on Learning Representations (ICLR)*, 2025.

2025: Payal Mohapatra*, **Akash Pandey***, Xiaoyuan Zhang*, and Qi Zhu. Can LLMs understand unvoiced speech? Exploring emg-to-text conversion with LLMs. *Association for Computational Linguistics (ACL)*, 2025.

2025: Payal Mohapatra, Yueyuan Sui, **Akash Pandey**, Stephen Xia, and Qi Zhu. MAESTRO : Adaptive sparse attention and robust learning for multimodal dynamic time series. *In Under review*, 2025.

2023: Akash Pandey, Wei Chen, and Sinan Keten. Mechanical property prediction and motif identification in spider silk material. *Society of Engineering Science (SES)*, 2023.

2023: Payal Mohapatra*, **Akash Pandey***, Yueyuan Sui*, and Qi Zhu. Effect of attention and self-supervised speech embeddings on non-semantic speech tasks. In *Proceedings of the 31st ACM International Conference on Multimedia*, pages 9511–9515, 2023.

2023: Payal Mohapatra*, **Akash Pandey***, Sinan Keten, Wei Chen, and Qi Zhu. Person identification with wearable sensing using missing feature encoding and multi-stage modality fusion. In *ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–2, 2023.

2022: Payal Mohapatra, **Akash Pandey**, Bashima Islam, and Qi Zhu. Speech disfluency detection with contextual representation and data distillation. In *Proceedings of the 1st ACM International Workshop on Intelligent Acoustic Systems and Applications*, IASA '22, page 19–24, 2022.

2016: Akash Pandey and Arockiarajan Arunachalakasi. Methods for measuring the life of the MFC as sensor and actuator at high temperature. In *Smart Materials, Adaptive Structures and Intelligent Systems*, volume 50480. American Society of Mechanical Engineers, 2016.

2016: Akash Pandey and A Arockiarajan. Experimental studies on fatigue behavior of macro fiber composite (MFC) under mechanical loading. In *Sensors and Smart Structures Technologies for Civil, Mechanical, and Aerospace Systems 2016*, volume 9803, pages 818–825. SPIE, 2016.

Teaching Assistantship

- **Winter 2025:** Scientific and Embedded Programming in Python, Northwestern University
- **Spring 2022:** Introduction to Aerospace Engineering, Northwestern University
- **Winter 2016:** Finite Element Analysis, IIT Madras

Services

Primary Reviewer: Neurips'25, npj Computational Materials, IEEE Transactions on Computational Biology and Bioinformatics

Journal Club: NICO Reading Group, Northwestern University

Mentoring

Current.....

Julia Levenshteyn (Senior, Lake Forest College)

Past.....

Xiaoyuan Zhang (Masters, Northwestern University)

Yueyuan Sui (Masters, Northwestern University)